



INTERNATIONAL UNIVERSITY OF GRAND-BASSAM  
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SCHOOL OF SCIENCE AND ENGINEERING

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# Requirements I — Why Users Cannot Tell You, and Who to Ask

*The most expensive defect there is, the four reasons nobody can simply state what they need, and how to find the stakeholder who is not in the room*

Session 7 of 28 • Tuesday 29 September 2026 • 1:00 – 2:20 PM

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## Requirements I — Why Users Cannot Tell You, and Who to Ask

SESSION 7 OF 28

Tuesday, 29 September 2026 • 1:00 – 2:20 PM • Unit 2 • Requirements

### LEARNING OBJECTIVES

- ✓ Explain why a wrong requirement is the most expensive class of defect, and trace one SANTE-BASSAM error through the life cycle to watch the cost multiply.
- ✓ Name the four reasons a stakeholder cannot simply tell you what they need, and the countermeasure for each.
- ✓ Identify the stakeholders of a system — including the ones who are not users — and place them on a power-interest grid.
- ✓ Choose an elicitation instrument for a given stakeholder and a given kind of information, and defend the choice.

### FROM UNIT 1

You finished Unit 1 holding a backlog of sixteen stories and a Definition of Done. Every one of those stories is a **hypothesis** about what somebody at the health post needs — and most of them were written by six

students who have met the clerk once. Unit 2 is how you find out which hypotheses are wrong, before sprint 2 spends a fortnight building them.

TEAM PROJECT · 0

first 15 minutes of class

## Plan sprint 2 — with WIP limits and the XP decision

*Applying Sessions 5 and 6. Sprint 2 is the tablet sprint, weeks 7–8, review Friday 30 October. This is the slot Session 5 promised you.*

### WHAT HAPPENS

1. **Layer 1 — the goal.** One sentence, a named person, a date, something observable at the post. Five minutes, whole team, Product Owner proposes.
2. **Layer 2 — the stories.** Pull from the top of the backlog to your measured velocity. Say what you left out and why. Four minutes.
3. **Layer 3 — tasks and the board.** Three pairs take one story each and break it down. Then put a **WIP limit** on each active column of your board and write one line justifying each number for six people. Six minutes.

### TWO DECISIONS TO RECORD WHILE YOU ARE ALL IN THE ROOM

#### The XP practices

which of the nine you are adopting this semester, and for each one you are not, what you will do instead (your HW06 paragraph, agreed as a team)

#### The DoD line

item 3, “demonstrated on the target device”, rewritten for a tablet in a village with no network

#### The review date

who goes to the post on 30 October, and who asks the nurse the one question a day in between

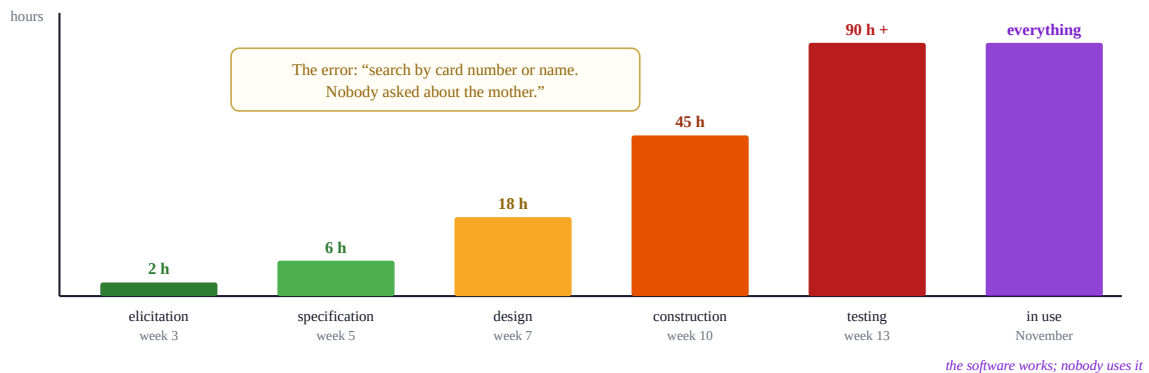
**Output:** sprint 2 in the progression log, with owners, WIP limits and the amended Definition of Done.

**Next session's slot:** the stakeholder map and who interviews whom — so bring today's backlog with you.

## 1 · The most expensive defect there is

A defect in code costs what it costs to fix the code. A defect in a *requirement* costs something else entirely, because it does not sit still. It propagates: into a design decision, then into the modules built on that

decision, then into tests that faithfully confirm the wrong behaviour, and finally into the habits of people who have arranged their work around it. Reversing it means unwinding every one of those.



**Figure 1.** One requirements error, priced at each phase it survives. Boehm's data, replicated many times since: roughly an order of magnitude per phase. The last bar is the one to look at — it is not a bigger number, it is a different kind of failure.

### THE UNCOMFORTABLE IMPLICATION

Most student projects fail not because the code is bad, but because the team built something nobody needed, *competently*. Your programming ability offers no protection against this whatsoever. Only contact with reality does — and contact with reality is a skill with techniques, which is what Unit 2 teaches.

## 2 · Why users cannot simply tell you

The naive model of requirements is transcription: the customer knows what they want, you write it down, you build it. It fails for four independent reasons. They are independent because each one needs a different countermeasure — asking harder does not fix any of them.

## Tacit knowledge

THEY KNOW IT BUT CANNOT SAY IT

Expertise becomes automatic and stops being available for description. The answer you get is “you just know”.

**Countermeasure:** observe, then ask about one concrete instance at a time. Never ask for the rule; ask about last Tuesday.

## Solution-first framing

THEY TELL YOU A DESIGN, NOT A NEED

People helpfully skip to what they imagine the software should do, and the underlying need is lost behind it.

**Countermeasure:** ask *why* until you reach something that is not about software, then generate two or three alternative solutions and let evidence choose.

## Unarticulated needs

SO OBVIOUS NOBODY MENTIONS IT

Everyone in the room shares an assumption, so nobody states it. These surface in week 12, as defects.

**Countermeasure:** read the documents and walk the physical environment before the first interview. Ask about failure: “what happens when the power goes off at 7 pm?”

## Conflict

TWO STAKEHOLDERS WANT INCOMPATIBLE THINGS

And usually neither knows it, because they have never been in the same room.

**Countermeasure:** write both statements down side by side with their sources, and escalate to someone with authority over both. The output is a requirement with a **named decider**.

## APPLIED TO SANTE-BASSAM

*The four failures, as they will actually happen to you at the post*

### Tacit — the nurse

**You hear:** “How do you decide who needs follow-up?” — “You just know.”

**It hides:** children under five with a second missed appointment; pregnant women past 32 weeks; anyone on TB treatment.

**You do:** sit beside her for one clinic morning. After each patient: “will you want to see this one again — when — how did you decide?” Write the rule down and show it to her. S-12’s acceptance criteria come from that piece of paper.

### Solution-first — the clerk

**You hear:** “I need a dropdown of the 23 villages.”

**It hides:** she misspells village names and the officer rejects the report. The need is *village names the report will accept*.

**You do:** ask why three times, then generate alternatives — dropdown, type-ahead, auto-correct, normalise at report time. Which one wins depends on how she actually types, and only observation tells you that.

### Unarticulated — everyone

**Nobody mentions:** a child brought by a neighbour must be registrable without the mother present; the tablet screen is sometimes the only light in the room.

**You do:** read the paper register and the Ministry guidelines *before* the first interview, then walk the failure checklist with the nurse.

### Conflict — clerk against officer

**You hear:** the clerk wants registration under ninety seconds; the officer wants seven more fields on every record.

**You do:** write both down with sources and take it to the nurse-in-charge, who has authority over both. Her ruling — required fields at registration, the rest at the first visit — is a requirement with a named decider, and it stops the argument recurring.

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## 3 · Who to ask: stakeholder analysis

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Primary users are one group among many, and they are rarely the group that stops your project. A stakeholder is anyone who **funds, approves, operates, maintains, regulates, supports, supplies data to, or is affected by** the system. Those eight verbs are the checklist — go through them one at a time rather than trying to think of people.

## THE POWER-INTEREST GRID

Place each stakeholder by two axes. **Power** — can they change or stop the project? **Interest** — how much does the outcome affect them? The quadrant then prescribes the engagement, which is the whole point of drawing it.

| Quadrant                  | Engagement                                          | What going wrong looks like                        |
|---------------------------|-----------------------------------------------------|----------------------------------------------------|
| High power, high interest | Manage closely — involve in every review            | They find out in week 12 and withdraw support      |
| High power, low interest  | Keep satisfied — brief, do not flood                | Ignored until they block the release               |
| Low power, high interest  | Keep informed — they are your best source of detail | Their real workflow never reaches the requirements |
| Low power, low interest   | Monitor — cheaply                                   | Rarely the problem                                 |

### The empty chair

The most dangerous stakeholder is the one who does not exist yet. On SANTE-BASSAM there is **no IT support at the health post after hand-over** — nobody owns backups, upgrades or a broken tablet in February. That absence is a stakeholder position, and it has already generated a requirement: maintainability by someone who did not write the code. When you draw a grid, always name the operational task nobody owns.

## APPLIED TO SANTE-BASSAM

Eleven stakeholders — and the two the team did not know about on day one

| Stakeholder              | Quadrant       | What they want                                                      | How you engage them                                                                               |
|--------------------------|----------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Nurse-in-charge          | manage closely | The follow-up list; nothing lost; a report she can sign her name to | Fortnightly review; one question a day by message; signs the acceptance tests                     |
| Front-desk clerk         | manage closely | Faster than paper; no double entry                                  | Two mornings of observation; tries every increment first. <b>Her reaction is the requirement.</b> |
| District health officer  | keep satisfied | Totals by village and outcome she can trust                         | One 45-minute interview in week 3; a mock report in week 9                                        |
| Community health workers | manage closely | Offline registration; a battery that survives the outreach day      | Shadow one full outreach day; they test sprint 2                                                  |
| Pharmacy assistant ★     | keep informed  | To see today's visit outcomes so she can dispense                   | Found by referral in week 4. One new story and a sync-latency constraint                          |
| Patients and guardians   | keep informed  | To be found at the next visit, card or no card; consent respected   | Not interviewed directly — represented by the nurse and by the register                           |
| Ministry of Health       | keep satisfied | The identifier format, the report template, retention rules         | Document analysis. It does not negotiate                                                          |
| Post supervisor          | keep satisfied | No disruption to the clinic                                         | Briefed before each visit; approves your presence                                                 |
| Your instructor          | keep satisfied | Evidence of engineering, not of typing                              | Stands in for absent stakeholders; marks the SRS                                                  |
| Future maintainers       | keep informed  | Code and documents they can pick up cold                            | Represented by the maintainability requirement, by nobody else                                    |

IT support ★

the  
empty  
chair

—

**Does not exist.** Name the tasks  
nobody owns after hand-over,  
and decide now who does them

## 4 · Which instrument, and in what order

Six instruments, and the order matters more than the choice. Two rules run the whole plan: **never ask what you could have read**, and **observe before you ask**, so that your questions are about things you have actually seen.

| Instrument                               | Best for                                             | Blind to                                                     | On SANTE-BASSAM                                                        |
|------------------------------------------|------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Document analysis</b>                 | Fields, formats, legal constraints, the vocabulary   | What people actually do with the documents                   | Paper register, patient card, Ministry template — <b>first, always</b> |
| <b>Observation</b><br>contextual inquiry | Tacit knowledge, workarounds, real frequencies       | Rare events; motives; anything in someone's head             | Two mornings at the desk; one outreach day with a CHW                  |
| <b>Interview</b>                         | Rationale, exceptions, history, disagreement         | Anything they cannot articulate — and it invites invention   | Nurse and officer, <i>after</i> the observation mornings               |
| <b>Prototype</b>                         | Reactions to something concrete; settling arguments  | Performance, volume, anything at scale                       | Paper screens for search and the Monday list                           |
| <b>Workshop</b>                          | Resolving a known conflict, with the decider present | Quiet stakeholders; anything politically costly to say aloud | Only if the clerk-versus-officer conflict does not resolve             |
| <b>Questionnaire</b>                     | Many respondents, known questions                    | Everything you did not already think to ask                  | Not used — eleven stakeholders, not eleven hundred                     |

### THE ORDERING RULE, AND WHY IT SAVES YOU A WEEK

Documents → observation → interviews → prototype. Reading the register first means you never waste an interview asking what fields exist. Observing next means your interview questions are about a notebook you have *seen*, which is a question people can answer, rather than about “your workflow”, which is a question nobody can.

### SESSION 7 TAKEAWAY

A wrong requirement is the only defect that gets more expensive every week you do not find it, and the last stage of that curve is not an expensive fix but working software nobody uses. Users cannot simply tell you what they need, for four reasons that each need a different countermeasure. So you go and look: name every stakeholder using the eight verbs, place them by power and interest, notice the empty chair — and then read before you watch, and watch before you ask.

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## 5 • In-class exercise • the stakeholders you did not think of

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Twelve minutes, in your three pairs, then three minutes comparing. **Not SANTE-BASSAM** — we use Project B from Session 4, the mobile-money savings app for an Abidjan start-up, so that you practise on a system you have no map for yet.

### Build the map (in pairs, 12 minutes)

1. **Eight stakeholders**, at least three of whom are not users. Walk the eight verbs one at a time — funds, approves, operates, maintains, regulates, supports, supplies data to, is affected by. Do not brainstorm; go through the list.
2. **Place each on the grid**, and name **the empty chair**: which operational task has nobody attached to it after launch?
3. **Two instruments**. For your two “manage closely” stakeholders: which instrument first, and what would you read before you met them?
4. **One conflict**. Two stakeholders who want incompatible things and do not know it. Who decides between them?

#### WHAT MOST GROUPS MISS

The central bank (the regulator), the mobile-network operator whose channel the app depends on, and the neighbourhood agent who converts cash. That last one is the pharmacy assistant of this project: invisible from the office, essential at the counter, and discovered only by asking somebody *who else should we talk to?*

## 6 · Homework · individual · due before Session 8 (Friday 2 October)

Yours alone, about forty-five minutes, none of it hard. It continues the exercise above and prepares the elicitation the team will run in weeks 3 and 4.

#### TEMPLATE

Download **CSC4305\_HW07\_Template.docx** from **CSC\_4305\_SWE-Assignments**, fill it in, **rename** it **CSC4305\_HW07\_FirstName\_LastName.docx** and **commit** it to **/homework/S07/** in **your own private repository**.

**A .docx file**. Not a PDF, not a link. Full instructions are in the *Read this first* box at the top of the template.

- Q1** **Finish the grid you started in class**. Your eight Project B stakeholders, placed, with one line each on how you would engage them. Mark the empty chair.

Three of the eight must be non-users. If all eight touch the app, you skipped some verbs.

- Q2** **One stakeholder nobody in the room named**. Add a ninth to the SANTE-BASSAM table in §3 — somebody genuinely not listed there. Say which verb found them, which quadrant they belong in, and what would go wrong if you never spoke to them.

**Q3 Diagnose four statements.** For each, say which of the four failures it is — tacit, solution-first, unarticulated or conflict — and give the one question or action you would use next.

(a) “Just make it work like the paper register.” (b) “We need an SMS reminder system.” (c) “I can tell straight away which ones will come back.” (d) The officer says every record needs a national ID; the clerk says half of her patients do not have one.

**Q4 Your two hours at the post.** You are given one clinic morning, alone. Write the plan: what you read beforehand, where you stand, what you count, and the three questions you will ask — none of which may be answerable by a document.

Half a page. The counting is the part students omit, and it is the part that turns an anecdote into a requirement.

**Q5 Document analysis, done for real.** Before Friday, read the paper-register template and the Ministry guideline in the course repository. List every field actually recorded, and flag the three you would not have guessed.

Bring this list to Session 8 — you will use it in the interview role-play.

**Q6 Reading.** Sommerville, *Software Engineering* 10th ed., §4.1–4.3.

**Where this goes next.** Friday is the craft: how to ask a question without inventing the answer, how to write down what you saw without silently deciding what it means, and how to get a reaction out of somebody who cannot describe what they want.

## Key terms

requirements elicitation

cost-of-correction curve

tacit knowledge

solution-first framing

unarticulated need

stakeholder

the eight verbs

power-interest grid

the empty chair

document analysis

contextual inquiry

named decider

## Readings and references

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- Mendelow, A. “Environmental Scanning: the Impact of the Stakeholder Concept”, ICIS, 1981 — the power–interest grid.
- Gottesdiener, E. *Requirements by Collaboration*, 2002.